

Voltage quality and economic activity: On-line Appendix

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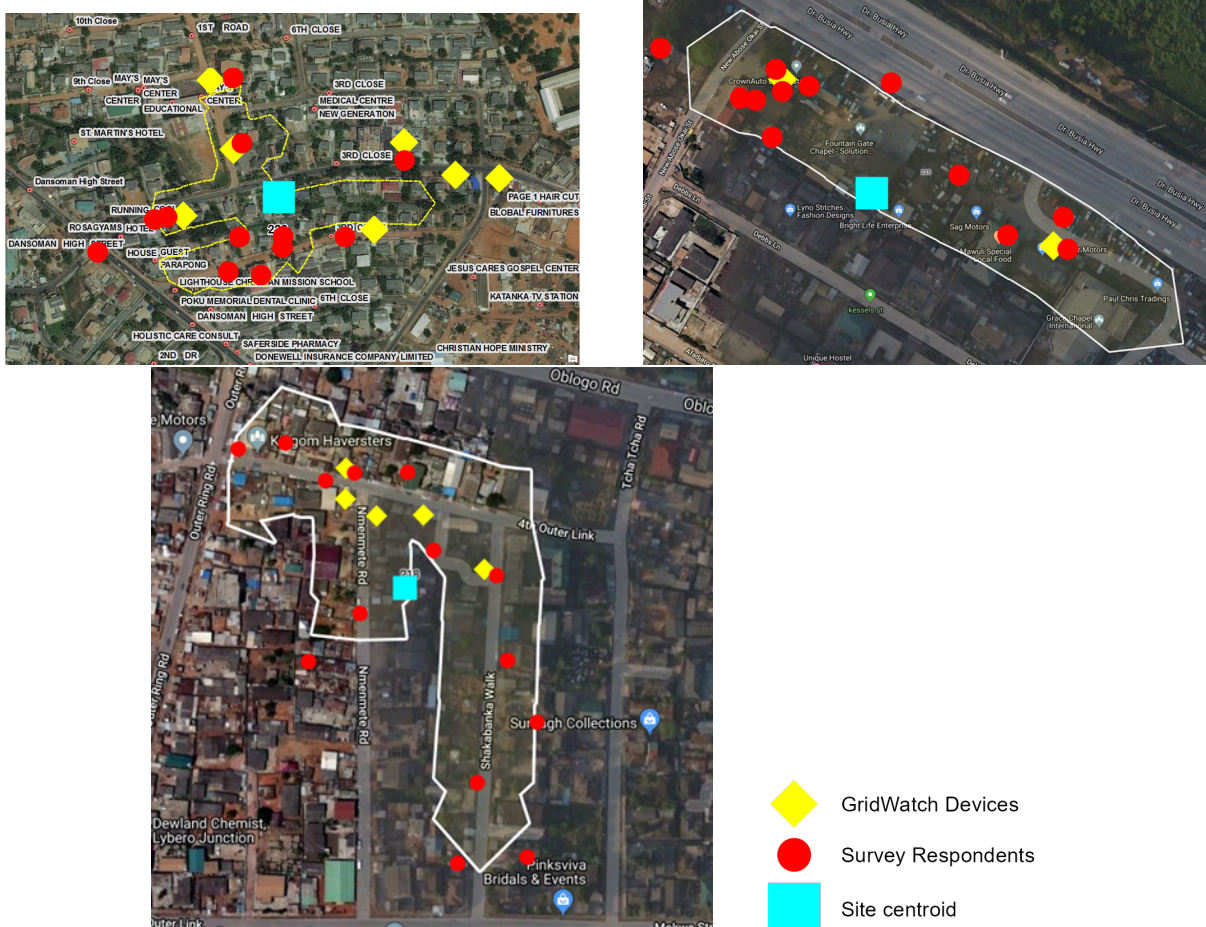
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Latest version available [here](#).

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Appendix C: Robustness checks

Figure C1: Examples of other sites



More examples of sites (as discussed in ??).

Table C1: Balance between panel and attrited respondents

	Matched N	Mean	Attrited N	Difference	p-val
<i>Respondent and Location</i>					
Age (years)	1575	39.23	426	2.73	0.000
Respondent is male	1575	0.35	426	0.02	0.547
Completed secondary education	1575	0.50	426	-0.05	0.050
Owens premises	1575	0.38	426	0.12	0.000
<i>Appliances</i>					
Any television (TV) at location	1575	0.71	426	-0.02	0.419
Any fridge at location	1575	0.62	426	0.01	0.835
Count of mobile phones	1575	2.17	426	0.05	0.605
Any electricity protective devices	1575	0.25	426	0.01	0.571
Count of reliability protective devices	1575	0.37	426	0.01	0.785
<i>Electricity and Energy</i>					
Pays someone else for electricity	1575	0.09	426	-0.08	0.000
Count of meter users	1566	1.84	424	-0.44	0.001
Last month electricity spending (USD)	1559	16.68	413	0.46	0.615
Has generator	1575	0.04	426	0.02	0.070
Count of alternative fuels used in past 3 months	1575	0.91	426	-0.07	0.107
Last month spending on alternative fuels (USD)	1575	6.24	426	-0.30	0.497
Reported total outage hours in past month	1575	39.88	426	3.44	0.151
Reported hours of bad voltage in past month	1566	47.49	422	4.29	0.408
Any burnt/broken appliances in past 12 months	1575	0.27	426	0.03	0.141
Amt. spent on burnt/broken apps in past year (USD)	1562	9.30	426	4.10	0.002
<i>Household Characteristics</i>					
Adults	746	2.39	251	0.27	0.002
Children (<18)	746	1.21	251	0.24	0.014
Last month HH income (USD)	714	352.97	234	-34.53	0.502
Share of HH adults with paid jobs in last week	746	0.66	251	-0.04	0.123
<i>Business Characteristics</i>					
Number of workers	829	1.97	175	-0.11	0.579
Last month business revenue (USD)	723	398.22	147	-65.65	0.292
Last month business costs (USD)	829	272.95	175	-29.26	0.449
Last month business profit (USD)	646	98.15	131	-31.02	0.063
Usual business open hours	829	12.07	175	0.20	0.397
Any non-electric business machines at location	829	0.09	175	0.00	0.992
Business engaged in retail activities (=1)	829	0.44	175	-0.07	0.085
Business engaged in manufacturing activities (=1)	829	0.20	175	0.11	0.000
Business engaged in other service activities (=1)	829	0.36	175	-0.03	0.409
Business activity likely using electricity (=1)	829	0.22	175	0.09	0.004

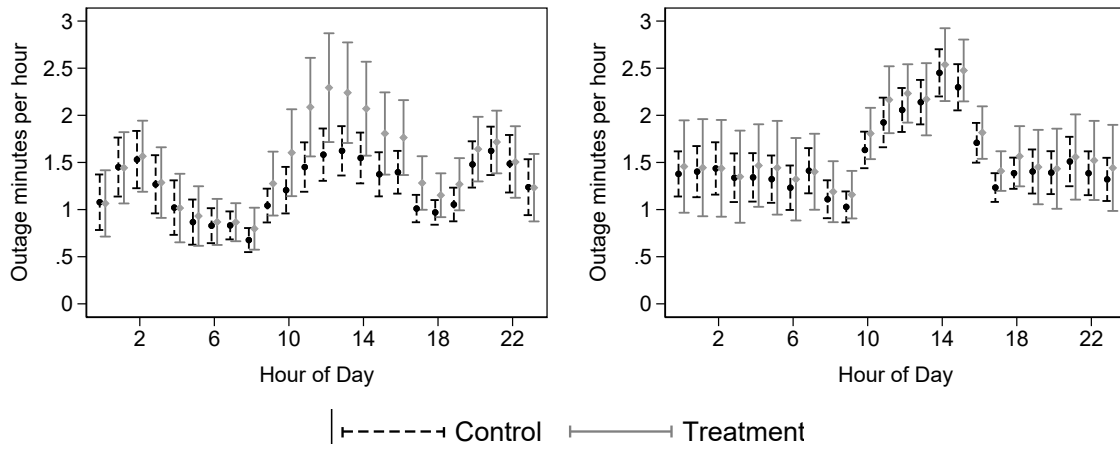
This table shows means in the baseline period for survey respondents, pooling businesses and households, and tests for significance of the differences in means by whether the respondent was also surveyed at the endline. The p-value for the joint F-test for household baseline characteristics is <0.001. The p-value for the joint F-test for business baseline characteristics is 0.024. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C2: Correlation between attrited respondents' characteristics and treatment status

	Mean	LB Treat	N
Age (years)	37.11 [12.40]	-1.20 (1.09)	426
Respondent is male	0.37 [0.48]	-0.07 (0.05)	426
Completed secondary education	0.54 [0.50]	0.04 (0.05)	426
Owns premises	0.31 [0.47]	-0.12*** (0.04)	426
Any television (TV) at location	0.69 [0.47]	0.08* (0.04)	426
Any fridge at location	0.60 [0.49]	0.02 (0.05)	426
Count of mobile phones	2.22 [2.00]	-0.20 (0.18)	426
Any electricity protective devices	0.26 [0.44]	-0.03 (0.04)	426
Count of reliability protective devices	0.37 [0.70]	-0.02 (0.07)	426
Pays someone else for electricity	0.17 [0.38]	-0.00 (0.04)	426
Count of meter users	2.35 [2.94]	-0.13 (0.24)	424
Last month electricity spending (USD)	17.12 [16.58]	-1.76 (1.64)	413
Has generator	0.02 [0.14]	0.01 (0.02)	426
Count of alternative fuels used in past 3 months	0.96 [0.82]	0.05 (0.08)	426
Last month spending on alternative fuels (USD)	6.21 [7.70]	0.63 (0.78)	426
Reported total outage hours in past month	35.39 [40.27]	2.05 (4.10)	426
Reported hours of bad voltage in past month	42.73 [102.40]	0.90 (9.27)	422
Any burnt/broken appliances from voltage fluctuations in past 12 months	0.24 [0.43]	-0.01 (0.04)	426
Amt. spent on burnt/broken apps in past year (USD)	4.59 [19.44]	1.19 (2.02)	426

This table shows the correlation between the attrited respondents' characteristics and treatment status. The sample is restricted to respondents who do not participate in the endline survey. We regress each respondent characteristic at baseline on a dummy variable equals one if the respondent was in a treatment site at baseline. Each row represents an outcome. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure C2: Impacts of transformer injection on outages by time of day
A) Before construction B) After construction



The figure shows mean minutes of power outages by hour of day separately for treatment and control sites. 95% confidence intervals around treatment means are clustered at the site level. Panel A shows means for the year prior to the start of the transformer construction period, and Panel B shows means for the year after the end of the construction period. ?? shows impacts on average voltage.

Table C3: Impact of transformer injection intervention on hourly average voltage, robustness to implementation issues

	(1) All sites	(2) Commissioned sites (SMEC)	(3) New tx confirmed sites	(4) All sites, IV new tx with treat	(5) All sites
During construction	0.76 (1.09)	0.79 (1.09)	0.65 (1.18)	0.42 (1.22)	0.79 (1.55)
Treat X During	2.38 (1.60)	2.55 (1.63)	3.61** (1.70)		1.95 (1.93)
Post construction	5.94*** (1.74)	5.95*** (1.74)	5.26*** (1.85)	5.26*** (1.95)	5.95*** (1.82)
Treat X Post	5.48** (2.48)	6.06** (2.49)	8.41*** (2.48)		5.46** (2.59)
New transformer X Post				7.13** (3.13)	
New transformer X During				3.24 (2.05)	
Observations	9866078	9723260	8815828	9866078	9866078
Pre-constr. ctl. mean	219.18	219.18	220.24	219.18	217.95
Hour of day FE	Y	Y	Y	Y	Y
Week of year FE	Y	Y	Y	Y	Y
Site FE	Y	Y	Y	Y	Y
Revised constr. period	N	N	N	N	Y

This table shows the difference-in-difference results for the impact of the transformer injection treatment on hourly average voltage levels measured by GridWatch devices in each site. Column 2 drops two sites where the construction manager SMEC indicated the new transformer was not commissioned successfully. Column 3 drops sites where our own construction monitoring activities indicated no new transformer was built in a treatment site or a new transformer was built in a control site. Column 4 instruments for observing a new transformer during the construction monitoring visits with site treatment assignment. Column 5 defines the construction period as July 1, 2020-December 31, 2020 instead of October 1, 2020-March 31, 2021, based on reported dates of transformer construction activity from the construction manager, SMEC. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C4: Impact of transformer injection intervention on hourly outage minutes, robustness to implementation issues

	(1)	(2)	(3)	(4)	(5)
	All sites	Commissioned sites (SMEC)	New tx confirmed sites	All sites, IV new tx with treat	All sites
During construction	0.21*** (0.07)	12.35*** (4.32)	9.80** (4.52)	12.91*** (4.86)	0.06 (5.05)
Treat X During	-0.06 (0.12)	-4.21 (7.11)	-2.42 (7.64)		-6.68 (7.07)
Post construction	-0.08 (0.08)	-4.78 (4.68)	-7.43 (4.73)	-3.21 (5.50)	-2.37 (4.66)
Treat X Post	-0.21 (0.13)	-14.06* (8.04)	-13.49 (8.53)		-13.63* (8.22)
New Trafo X Post				-16.70 (10.37)	
New Trafo X During				-5.39 (9.07)	
Observations	10033086	9888612	8962703	10033086	10033086
Pre-constr.ctl. mean	1.39	83.40	84.60	83.40	89.00
Hour of day FE	Y	Y	Y	Y	Y
Week of year FE	Y	Y	Y	Y	Y
Site FE	Y	Y	Y	Y	Y
Revised constr. period		N	N	N	Y

This table shows the difference-in-difference results for the impact of the transformer injection treatment on hourly power outage minutes measured by GridWatch devices in each site. Column 2 drops two sites where the construction manager SMEC indicated the new transformer was not commissioned successfully. Column 3 drops sites where our own construction monitoring activities indicated no new transformer was built in a treatment site or a new transformer was built in a control site. Column 4 instruments for observing a new transformer during the construction monitoring visits with site treatment assignment. Column 5 defines the construction period as July 1, 2020-December 31, 2020 instead of October 1, 2020-March 31, 2021, based on reported dates of transformer construction activity from the construction manager, SMEC. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C5: Impacts of transformer injection intervention on additional voltage quality measures, hourly data

	N	Control Mean (SD)	Post (SE)	Treat × Post (SE)
Mean voltage during hour	9866078	219.18 (22.39)	5.94*** (1.74)	5.48** (2.48)
Absolute deviation from nominal	9866078	16.51 (18.60)	-3.74*** (1.14)	-0.83 (1.70)
Any voltage >20% below nominal	10033086	0.08 (0.26)	-0.03** (0.01)	-0.04** (0.02)
Minutes voltage >20% above nominal	10033086	0.00 (0.31)	-0.00 (0.00)	-0.00 (0.01)
Minutes voltage 10-20% below nominal	10033086	9.60 (19.83)	-2.96*** (0.93)	-4.21*** (1.41)
Minutes voltage >20% below nominal	10033086	2.76 (11.59)	-1.54*** (0.57)	-1.40* (0.85)
Minutes with no power (outage)	10033086	1.39 (8.59)	-0.08 (0.08)	-0.21 (0.13)

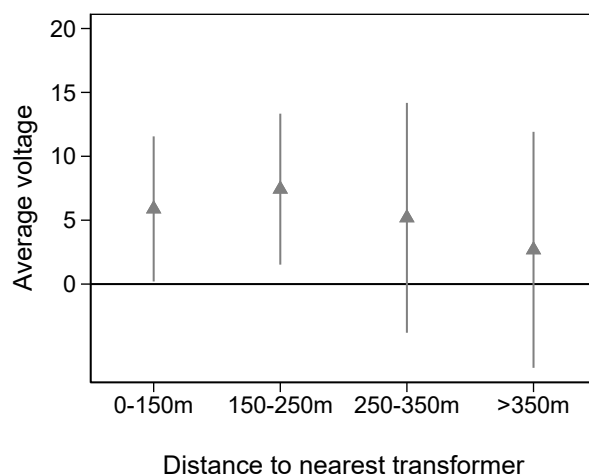
This table shows the difference-in-differences effects of the transformer injection intervention on measures of voltage quality using hourly data at the GridWatch device level. The minutes variables indicate the number of minutes in each hourly observation that the electricity had a certain status. “Any voltage >20% below nominal” is a dummy variable for whether voltage fell below this threshold at any point during an hourly observation. All regressions include hour of day, week of year, and site fixed effects. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C6: Impacts of transformer injection intervention on voltage quality, monthly data

	N	Control Mean (SD)	Post (SE)	Treat × Post (SE)
Hours with voltage >20% above nominal	14213	0.03 (0.47)	-0.01 (0.01)	-0.05 (0.06)
Hours with voltage 10-20% below nominal	14213	114.27 (152.16)	-33.31*** (11.24)	-52.00*** (17.15)
Hours with voltage >20% below nominal	14213	32.81 (99.40)	-18.18** (7.11)	-16.28 (10.58)
Hours with no power (outages)	14213	13.52 (13.60)	-0.45 (0.73)	-2.12 (1.40)
Number of spells with voltage <207	14213	251.92 (300.46)	-68.51*** (20.10)	-67.83** (31.18)
Number of spells with min voltage >200	14213	212.82 (255.67)	-60.65*** (16.55)	-50.46* (25.88)
Number of spells with min voltage btwn 184-200	14213	29.88 (46.22)	-6.15* (3.67)	-12.80** (5.19)
Number of spells with min voltage <184	14213	9.22 (14.45)	-1.70* (1.02)	-4.57*** (1.67)
Total duration of spells with voltage <207	14213	125.07 (181.45)	-43.46*** (14.11)	-54.68*** (20.80)
Total duration of spells with min voltage >200	14213	15.88 (18.97)	-3.67*** (1.25)	-3.98** (1.89)
Total duration of spells with min voltage btwn 184-200	14213	32.51 (45.81)	-7.76** (3.07)	-8.71** (4.23)
Total duration of spells with min voltage <184	14213	76.69 (154.91)	-32.03*** (11.31)	-41.99** (16.82)
Share of low-voltage time in spells with min voltage <184	11975	0.37 (0.38)	0.03 (0.02)	0.02 (0.04)
Mean spell length (hours)	11975	0.74 (2.19)	-0.29* (0.17)	-0.08 (0.23)
Median spell length (hours)	11975	0.12 (0.58)	-0.05 (0.04)	0.06 (0.05)
Maximum spell length (hours)	11975	11.98 (22.38)	-3.42** (1.72)	-5.90** (2.65)
Mean of mean voltage during a spell	11975	189.84 (35.33)	-8.33*** (2.50)	-15.76*** (4.09)
Median of mean voltage during a spell	11975	191.94 (36.10)	-7.95*** (2.54)	-15.70*** (4.10)
Mean of minimum voltage during a spell	11975	187.41 (36.69)	-8.54*** (2.63)	-15.32*** (4.30)

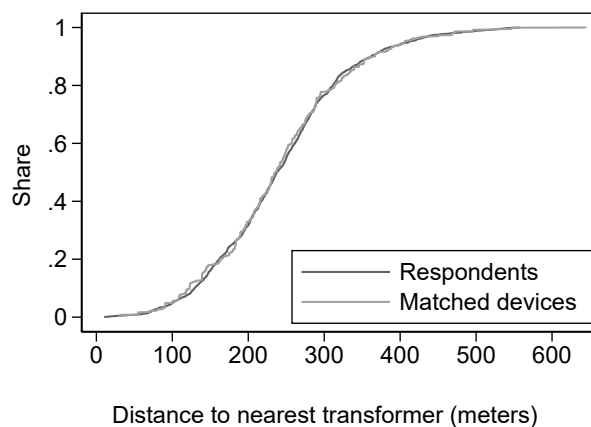
This table shows the difference-in-differences effects of the transformer injection intervention on measures of voltage quality using monthly data at the GridWatch device level. Outcomes in all but the first row are measured based on identifying individual low-voltage “spells” during which voltage fell below 207V (10% below nominal) in any 2-minute interval. Individual spells with different characteristics are then aggregated to the month-device level. Months where there were no low-voltage spells for particular devices are assigned a 0 for outcomes that are not conditional on experiencing at least one such spell. Number of spells refers to the number of individual low-voltage spells in a device-month. Total hours of spells take the sum of the duration of individual spells in a device-month. Mean, median, and maximum spell length are statistics calculated over all individual spells in a device-month. Mean and median of mean spell voltage are statistics calculated over the mean voltage level within a spell for all individual spells in a device-month. Mean minimum voltage is calculated similarly. These statistics are conditional on any low-voltage spell being observed in a device-month. All regressions include month and site fixed effects. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure C3: Impact of transformer injection intervention by distance to the nearest transformer at baseline



The figure shows coefficients for the differential impacts on average voltage of baseline distance of the GridWatch device from the nearest transformer in treatment sites relative to control sites in the post-construction period.

Figure C4: Distribution of distances between respondents and devices to the nearest transformer



Distance to the nearest transformer is measured during the baseline period before new transformers were added in treatment sites. The sample of devices includes only those that are matched to respondents as being the closest device.

Table C7: Impact of transformer injection intervention on primary outcomes for businesses

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post $\times$ Treat (SE)
Voltage damage and protection index	1658	-0.15 (0.93)	-0.07 (0.06)	0.13 (0.08)	-0.08 (0.08)
Any voltage-related damage, last 12 months (=1)	1658	0.22 (0.41)	-0.02 (0.03)	0.04 (0.04)	-0.05 (0.04)
Any reliability protective device owned (=1)	1658	0.19 (0.39)	-0.03*** (0.01)	0.04 (0.03)	-0.01 (0.02)
Amt. spent on burnt/broken apps in past year (USD)	1628	7.18 (27.84)	-3.80** (1.62)	-0.28 (2.05)	1.66 (2.21)
Value of protective devices owned (USD)	1426	5.14 (27.95)	-3.10** (1.53)	-0.55 (2.25)	-0.23 (2.22)
Reported hours of bad voltage in past month	1636	40.61 (82.68)	-39.84*** (4.38)	8.28 (7.83)	-10.11 (8.38)
Reported total outage hours in past month	1619	31.99 (31.34)	-29.10*** (2.28)	-0.40 (3.01)	1.31 (3.07)
WTP for perfect reliability and quality (USD)	1658	3.10 (4.35)	-1.07*** (0.25)	-0.26 (0.30)	0.33 (0.35)
WTP for perfect voltage and half outage hours (USD)	1658	1.58 (2.78)	-0.26 (0.19)	-0.27 (0.20)	0.34 (0.25)
WTP for no outages and half bad voltage hours (USD)	440	1.88 (3.46)	0.34 (0.41)	-0.26 (0.43)	0.88 (0.59)
Total no. of appliances owned	1658	7.07 (5.54)	-0.20 (0.12)	-0.02 (0.38)	0.20 (0.18)
Any alt. energy source used in last month (=1)	1658	0.06 (0.24)	-0.01 (0.01)	-0.00 (0.02)	0.00 (0.01)
Last month electricity spending (USD)	1594	17.63 (16.99)	-3.92*** (0.73)	-1.74 (1.22)	0.21 (0.96)

This table shows the difference-in-difference results from the ?? for businesses only. Each row represents an outcome. All variables measuring values are in USD. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Refer to [Table D5](#) for impacts of transformer injection on additional business outcomes.

Table C8: Impact of transformer injection intervention on primary outcomes

		Control					Post	
	N	Mean (SD)	Post (SE)	FDR q-val	Treat (SE)	FDR q-val	× Treat (SE)	FDR q-val
Voltage damage and protection index	3150	0.00 (1.00)	-0.11** (0.05)	0.042	0.09 (0.06)	0.678	-0.10* (0.06)	0.331
Any voltage-related damage, last 12 months (=1)	3150	0.25 (0.43)	-0.05* (0.03)	0.122	0.04 (0.03)	0.678	-0.05 (0.04)	0.418
Any reliability protective device owned (=1)	3150	0.25 (0.44)	-0.02** (0.01)	0.030	0.02 (0.02)	0.798	-0.02 (0.01)	0.331
Amt. spent on burnt/broken apps in past year (USD)	3080	9.28 (33.80)	-5.77*** (1.37)	0.001	0.05 (1.77)	0.988	1.23 (1.91)	0.687
Value of protective devices owned (USD)	2668	5.49 (23.16)	-2.32** (0.92)	0.030	0.36 (1.42)	0.975	0.32 (1.63)	0.899
Reported hours of bad voltage in past month	3110	43.18 (87.51)	-42.48*** (4.72)	0.001	7.60 (7.41)	0.678	-9.09 (7.70)	0.453
Reported total outage hours in past month	3081	32.15 (31.08)	-29.32*** (2.05)	0.001	1.69 (2.60)	0.798	-1.18 (2.74)	0.811
WTP for perfect reliability and quality (USD)	3150	3.29 (4.41)	-1.41*** (0.21)	0.001	-0.34 (0.25)	0.678	0.44 (0.28)	0.331
WTP for perfect voltage and half outage hours (USD)	3150	1.58 (2.71)	-0.37** (0.16)	0.042	-0.17 (0.17)	0.678	0.21 (0.21)	0.549
WTP for no outages and half bad voltage hours (USD)	850	1.95 (3.34)	-0.07 (0.30)	0.821	-0.37 (0.32)	0.678	0.99** (0.44)	0.331
Total no. of appliances owned	3150	8.59 (5.98)	-0.04 (0.08)	0.719	-0.05 (0.35)	0.988	0.08 (0.13)	0.687
Any alt. energy source used in last month (=1)	3150	0.05 (0.22)	-0.01 (0.01)	0.416	0.01 (0.01)	0.896	-0.00 (0.01)	0.933
Last month electricity spending (USD)	3050	17.72 (16.95)	-3.92*** (0.60)	0.001	-1.90* (1.02)	0.678	0.53 (0.80)	0.687
Last month business profit (USD)	1104	98.60 (143.83)	-6.77 (10.50)	0.590	-12.44 (11.53)	0.678	3.55 (13.67)	0.899
Last month business revenue (USD)	1280	396.21 (625.91)	84.40* (44.75)	0.095	-0.80 (51.71)	0.988	-92.58 (57.75)	0.331
Last month business costs (USD)	1206	276.34 (358.93)	82.04** (36.68)	0.046	25.30 (37.84)	0.798	-98.48** (49.45)	0.331
Last month HH income (USD)	1358	332.33 (473.05)	-34.24 (36.83)	0.430	16.93 (41.60)	0.896	-75.43 (49.79)	0.331

This table shows the difference-in-difference results from the ?? pooling businesses and households. Each row represents an outcome. All variables measuring values are in USD. Results are qualitatively unchanged when using logged versions of continuous outcomes. Sample sizes vary for some questions because of missing data, particularly when respondents were unable to estimate monetary values with a high degree of confidence, or because some questions were only asked to a subset of respondents. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is a household or a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Sharpened FDR q-values following Anderson (2008) are also shown.

Table C9: Impact of transformer injection intervention on primary outcomes, accounting for implementation issues and construction timing

	All sites	Comm- issioned sites (SMEC)	New transformer confirmed sites	All sites, IV new tx with treat	No movers	All sites, except close sites
Voltage damage and protection index	-0.10* (0.06)	-0.11* (0.06)	-0.13** (0.07)	-0.14* (0.08)	-0.11* (0.06)	-0.16** (0.08)
Any voltage-related damage, last 12 months (=1)	-0.05 (0.04)	-0.06 (0.04)	-0.07* (0.04)	-0.06 (0.05)	-0.06 (0.04)	-0.08* (0.05)
Any reliability protective device owned (=1)	-0.02 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.02 (0.02)	-0.01 (0.01)	-0.02 (0.02)
Amt. spent on burnt/broken apps in past year (USD)	1.23 (1.90)	1.09 (1.93)	0.21 (2.04)	1.63 (2.53)	0.88 (1.89)	1.49 (2.64)
Value of protective devices owned (USD)	0.32 (1.62)	0.25 (1.65)	1.05 (1.82)	0.43 (2.16)	0.41 (1.72)	-0.85 (1.65)
Reported hours of bad voltage in past month	-9.09 (7.69)	-10.19 (7.80)	-12.79 (7.92)	-12.01 (10.08)	-13.26* (7.16)	-4.85 (8.42)
Reported total outage hours in past month	-1.22 (2.73)	-1.32 (2.76)	-2.01 (2.87)	-1.62 (3.59)	-1.16 (2.84)	1.84 (3.41)
WTP for perfect reliability and quality (USD)	0.44 (0.28)	0.46 (0.29)	0.54* (0.31)	0.58 (0.37)	0.36 (0.30)	0.23 (0.35)
WTP for perfect voltage and half outage hours (USD)	0.21 (0.21)	0.20 (0.21)	0.26 (0.23)	0.27 (0.27)	0.14 (0.21)	-0.00 (0.28)
WTP for no outages and half bad voltage hours (USD)	0.99** (0.44)	1.07** (0.44)	1.14** (0.48)	1.31** (0.58)	0.89** (0.43)	0.70 (0.62)
Total no. of appliances owned	0.08 (0.13)	0.09 (0.13)	0.14 (0.14)	0.11 (0.17)	0.11 (0.13)	0.14 (0.15)
Any alt. energy source used in last month (=1)	-0.00 (0.01)	-0.00 (0.01)	0.00 (0.01)	-0.00 (0.01)	0.00 (0.01)	0.02 (0.02)
Last month electricity spending (USD)	0.53 (0.80)	0.51 (0.81)	0.49 (0.87)	0.70 (1.06)	0.63 (0.83)	0.72 (1.00)
Last month business profit (USD)	3.67 (13.59)	4.16 (13.74)	10.11 (14.69)	4.97 (18.20)	5.20 (14.44)	6.28 (18.27)
Last month business revenue (USD)	-91.60 (57.48)	-90.49 (58.28)	-23.30 (57.11)	-125.79 (82.61)	-110.70* (60.84)	-103.81 (70.22)
Last month business costs (USD)	-97.46** (49.15)	-99.05** (49.64)	-54.44 (49.75)	-129.75* (68.16)	-105.28** (51.70)	-140.71** (59.55)
Last month HH income (USD)	-75.62 (49.63)	-82.31 (50.19)	-85.61 (53.64)	-102.29 (66.69)	-71.11 (52.38)	-53.05 (57.50)

This table shows the same difference-in-difference analyses presented in ???. Column 1 replicates the “Post $\times$ Treat” column from ??. Column 2 drops two sites in Kaneshie where the construction manager SMEC indicated the new transformer was not commissioned successfully. Column 3 drops sites where our own construction monitoring activities indicated no new transformer was built in a treatment site or a new transformer was built in a control site. Column 4 instruments for observing a new transformer during the construction monitoring visits with site treatment assignment. Column 5 drops respondents who moved between the baseline and endline survey. Column 6 drops control sites that are within 1.3 km from a treatment site (the median distance). Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C10: Correlations between adjusted voltage quality and primary outcomes

	N	Control Mean (SD)	Average voltage (SE)	Hours >10% below (SE)
Reported hours of bad voltage in past month	3111	24.08 (69.46)	-0.690*** (0.236)	0.150** (0.060)
Reported total outage hours in past month	3120	21.25 (38.40)	-0.311*** (0.111)	0.032* (0.018)
WTP for perfect reliability and quality (USD)	3131	2.52 (3.55)	-0.021*** (0.006)	0.005*** (0.002)
WTP for perfect voltage and half outage hours (USD)	3131	1.36 (2.35)	-0.010*** (0.003)	0.003*** (0.001)
WTP for no outages and half bad voltage hours (USD)	847	1.90 (3.17)	-0.005 (0.007)	0.002 (0.002)
Voltage damage and protection index	3131	-0.02 (0.97)	-0.002 (0.002)	0.001 (0.001)
Any voltage-related damage, last 12 months (=1)	3131	0.24 (0.42)	-0.001** (0.001)	0.000 (0.000)
Amt. spent on burnt/broken apps in past year (USD)	3063	6.94 (26.05)	-0.117*** (0.033)	0.028*** (0.010)
Any reliability protective device owned (=1)	3131	0.25 (0.43)	0.000 (0.001)	0.000 (0.000)
Value of protective devices owned (USD)	2649	4.82 (23.84)	-0.038 (0.037)	0.019 (0.015)
Any alt. energy source used in last month (=1)	3131	0.05 (0.21)	0.000 (0.000)	0.000 (0.000)
Total no. of appliances owned	3131	8.59 (5.73)	0.008 (0.007)	0.001 (0.002)
Last month electricity spending (USD)	3031	15.06 (13.56)	-0.016 (0.017)	0.002 (0.004)
Last month business profit (USD)	1103	89.57 (128.51)	0.300 (0.221)	-0.041 (0.038)
Last month business revenue (USD)	1273	404.66 (615.80)	1.677* (0.908)	-0.141 (0.169)
Last month business costs (USD)	1201	304.01 (446.26)	0.765 (0.633)	-0.066 (0.111)
Last month HH income (USD)	1348	301.69 (426.69)	-0.150 (0.663)	-0.056 (0.102)

This table shows the results from separate regressions of primary outcomes on measures of voltage quality. Each row represents a different outcome pooling business and household respondents. Profit is measured by directly asking the respondent, rather than by subtracting costs from revenues. Total reported costs are the sum of costs for specific items/activities and are not comprehensive. The columns indicate measures of voltage quality—the independent variables. Voltage is measured by assigning each respondent GridWatch data based on the nearest devices for either the last 30 days from the survey date or for the full baseline period (prior to November 1, 2020) and endline period (from April 1, 2021 - July 20, 2022). Voltage data is not available for respondents in one site where no devices were deployed. Mean voltage in control sites is 219.5V at baseline and 224.6V at endline. In all the regressions, we also control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the location includes both a household and a business, and district fixed effects. Standard errors are clustered at the site level. ?? shows the same using data adjusted for the device and the respondents' distances to transformer. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Appendix D: Pre-specified analyses of socioeconomic outcomes

Table D1: Impact of transformer injection intervention on willingness to pay outcomes

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post x Treat (SE)
WTP for perfect reliability and quality (USD)	3150	3.29 [4.41]	-1.41*** (0.21)	-0.34 (0.25)	0.44 (0.28)
WTP for perfect voltage and half outage hours (USD)	3150	1.58 [2.71]	-0.37** (0.16)	-0.17 (0.17)	0.21 (0.21)
Max monthly WTP for 1 unnan. 8hr outage	240	0.97 [1.72]	-0.51* (0.27)	-0.07 (0.30)	-0.17 (0.35)
Max monthly WTP for 1 announ. 8hr outage	262	1.27 [2.08]	-0.77*** (0.28)	0.10 (0.44)	0.12 (0.48)
Max monthly WTP for 4 unnan. 2hr outages	244	1.38 [2.20]	-0.84** (0.34)	-0.39 (0.40)	0.25 (0.43)
WTP for no outages and half bad voltage hours (USD)	850	1.95 [3.34]	-0.07 (0.30)	-0.37 (0.32)	0.99** (0.44)
Max WTP for generator	2724	317.00 [403.13]	-14.43 (24.44)	33.76 (23.80)	-38.35 (31.95)

Additional results from ???. All variables measuring values are in USD. Results are qualitatively unchanged when using logs. Sample sizes are lower for reliability scenarios that were only presented to a random subset of respondents. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D2: Impact of transformer injection intervention on alternative energy and defensive investment outcomes

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post x Treat (SE)
Any alt. energy source used in last month (=1)	3150	0.05 [0.22]	-0.01 (0.01)	0.00 (0.01)	-0.00 (0.01)
Value of alter. energy sources	3036	191.85 [3700.01]	-43.46 (29.54)	525.87 (660.07)	-615.26 (647.22)
Count of reliability protective devices	3150	0.35 [0.70]	-0.05*** (0.01)	-0.01 (0.04)	0.01 (0.02)
Value of protective devices owned (USD)	2668	5.49 [23.16]	-2.09** (0.86)	0.53 (1.44)	0.29 (1.63)
Has multi-phase system	2516	0.04 [0.21]	0.01* (0.01)	0.01 (0.01)	-0.01 (0.01)
Frequency of switching phases (z-score)	3150	0.02 [1.18]	-0.02 (0.04)	0.11 (0.08)	-0.08 (0.07)
Cost of installing phase system	2380	10.22 [116.78]	-3.58 (5.03)	2.10 (7.42)	-5.69 (7.01)

Additional results from ???. All variables measuring values are in USD. Results are qualitatively unchanged when using logs. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D3: Impact of transformer injection intervention on energy/electricity spending and burnt appliance outcomes

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post x Treat (SE)
Last month electricity spending (USD)	3050	17.72 [16.95]	-3.76*** (0.60)	-1.76 (1.08)	0.52 (0.80)
Has generator	3150	0.04 [0.20]	-0.01 (0.01)	-0.00 (0.01)	0.01 (0.01)
Generator fuel and maintenance costs in past 3 months	3150	5.15 [48.99]	-3.92*** (1.41)	-2.35 (1.96)	1.60 (1.66)
Has solar panels	3150	0.01 [0.08]	0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)
Amount spent on solar panel repairs in past 3 months	3140	0.52 [14.41]	-0.52 (0.52)	-0.52 (0.52)	0.52 (0.52)
Count of alternative fuels used in past 3 months	3150	0.92 [0.85]	0.07** (0.03)	-0.00 (0.05)	0.01 (0.04)
Last month spending on alternative fuels (USD)	3150	7.93 [28.53]	0.52 (1.00)	-1.27 (1.12)	0.74 (1.12)
Amount spent on charcoal as alt. fuel in past month	2850	12.16 [26.31]	13.29*** (3.23)	0.33 (1.63)	-3.10 (3.76)
Amount spent on gas as alt. fuel in past month	2852	26.45 [38.63]	16.35*** (1.98)	-0.91 (2.40)	-0.58 (4.26)
Amount spent on wood as alt. fuel in past month	3130	5.44 [144.76]	-4.26 (4.46)	-5.36 (5.23)	5.62 (4.53)
Total hrs per day lightbulbs are on	3150	9.17 [5.68]	-0.90*** (0.34)	0.19 (0.39)	-0.58 (0.47)
Any voltage-related damage, last 12 months (=1)	3150	0.25 [0.43]	-0.04 (0.03)	0.04 (0.03)	-0.05 (0.04)
Has burnt/broken apps. that were not replaced in past year	3132	0.08 [0.28]	-0.02* (0.01)	0.02 (0.02)	-0.01 (0.02)
Amt. spent on burnt/broken apps in past year (USD)	3080	9.28 [33.80]	-5.56*** (1.36)	0.18 (1.77)	1.23 (1.90)

Additional results from ???. All variables measuring values are in USD. Results are qualitatively unchanged when using logs. Sample sizes are lower for reliability scenarios that were only presented to a random subset of respondents. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D4: Impact of transformer injection intervention on electricity-related indices

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post x Treat (SE)
Outage backup power index	3150	0.00 [1.00]	-0.06 (0.04)	0.02 (0.06)	-0.01 (0.04)
Freq. of wetcell batt./generator use during outage (normalized)	3150	0.06 [1.26]	-0.06 (0.04)	-0.00 (0.07)	0.02 (0.05)
Share of apps. using solar/generator during outage (normalized)	3114	0.10 [1.41]	-0.10 (0.06)	0.04 (0.09)	-0.04 (0.08)
Alternative energy/fuel sources index	3150	-0.00 [1.00]	-0.02 (0.05)	0.03 (0.07)	-0.03 (0.07)
Any alt. energy source used in last month (=1)	3150	0.05 [0.22]	-0.01 (0.01)	0.00 (0.01)	-0.00 (0.01)
Count of alt. light sources (normalized)	3150	0.10 [1.10]	-0.10 (0.06)	0.05 (0.09)	-0.05 (0.09)
Count of alt. fuel sources (normalized)	3150	-0.08 [0.96]	0.08** (0.03)	-0.00 (0.05)	0.01 (0.05)
Appliance protection index	3150	-0.00 [1.00]	-0.18*** (0.03)	-0.04 (0.05)	0.07* (0.04)
Count of reliability defensive apps. (normalized)	3150	0.08 [1.11]	-0.08*** (0.02)	-0.02 (0.06)	0.02 (0.03)
Has multi-phase system	2516	0.04 [0.21]	0.01* (0.01)	0.01 (0.01)	-0.01 (0.01)
Share of TVs plugged to TV guard (normalized)	110	2.54 [0.77]	-1.31*** (0.27)	-0.36 (0.24)	0.16 (0.39)
Share of fridges plugged to fridge guard (normalized)	232	1.35 [0.45]	-0.47*** (0.14)	-0.09 (0.10)	0.16 (0.20)

Additional results from ???. The main outcomes are indices; we also show results for the index components for completeness. Indices are constructed as the sum of normalized components, and are then normalized to have mean 0 and SD 1 for control respondents in the baseline. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table D5: Impact of transformer injection intervention on additional business outcomes

	N	Control Mean (SD)	Post (SE)	Treat (SE)	Post x Treat (SE)
Temporary business response type index	1658	0.00 [1.00]	-0.10 (0.06)	0.03 (0.09)	0.07 (0.10)
Temporary switch to alternative energy due to reliability (0-2)	1646	0.06 [0.29]	0.04*** (0.02)	0.02 (0.02)	-0.01 (0.03)
Temporary stop working, work less due to reliability (0-2)	1646	0.42 [0.71]	-0.03 (0.04)	0.02 (0.06)	-0.01 (0.06)
Temporary postpone working, woke same due to reliability (0-2)	1646	0.33 [0.63]	-0.13*** (0.03)	-0.02 (0.05)	-0.01 (0.05)
Temporary switch tools/labor due to reliability (0-2)	1646	0.16 [0.44]	0.01 (0.03)	-0.01 (0.03)	0.03 (0.04)
Temporary switch business activities due to reliability (0-2)	1646	0.15 [0.41]	-0.03 (0.03)	0.02 (0.03)	0.04 (0.04)
Temporary reduce labor due to reliability (0-2)	1646	0.20 [0.49]	-0.05 (0.04)	0.01 (0.04)	0.02 (0.05)
Other temporary response due to reliability	1646	0.02 [0.17]	-0.02* (0.01)	0.00 (0.01)	0.02 (0.02)
Temporary business response intensity index	1658	-0.00 [1.00]	-0.33*** (0.05)	0.61 (0.41)	-0.61 (0.41)
Days of switching to solar energy	1658	0.00 [0.00]	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Days of switching to generator	1658	0.01 [0.07]	-0.01* (0.00)	0.04 (0.04)	-0.05 (0.04)
Days of switching to wetcell	1658	0.00 [0.01]	-0.00 (0.00)	0.01 (0.01)	-0.01 (0.01)
Days stopping or postponing work in past 1 month	1658	0.00 [1.00]	-0.32*** (0.05)	0.02 (0.07)	-0.01 (0.07)
Percentage of business hours stopping work	1658	-0.00 [1.00]	-0.34*** (0.06)	0.09 (0.09)	-0.07 (0.09)
Permanent business response index	1658	0.00 [1.00]	-0.11 (0.07)	-0.09 (0.07)	0.27** (0.12)
Permanently substitute to non-electric tools or machines (0-1)	1646	0.02 [0.13]	-0.01 (0.01)	-0.01** (0.01)	0.01 (0.01)
Permanently substitute to more labor (0-1)	1646	0.01 [0.09]	-0.00 (0.00)	0.00 (0.01)	0.02 (0.01)
Permanently change industry/business (0-1)	1646	0.01 [0.12]	-0.00 (0.01)	-0.01 (0.01)	0.03** (0.01)
Purchase generator (0-1)	1646	0.01 [0.12]	-0.01 (0.01)	-0.01 (0.01)	0.01 (0.01)
Business qualitative assessments index	1658	0.00 [1.00]	-0.76*** (0.07)	0.00 (0.08)	-0.06 (0.09)
Perceived safety in area (1-5)	1656	3.30 [1.05]	0.00 (0.08)	-0.06 (0.10)	0.16 (0.11)
Importance of electricity as obstacle to business (1-5)	1646	3.90 [0.98]	-0.23*** (0.08)	0.01 (0.08)	-0.06 (0.11)
Belief that Dumsor is back (1-5)	1646	2.84 [1.31]	1.50*** (0.10)	0.14 (0.11)	-0.06 (0.12)
Expected reliability one year from today (1-3)	1078	2.34 [0.83]	0.35*** (0.08)	-0.00 (0.09)	0.05 (0.10)
Importance of finance/access to credit as a business obstacle (1-5)	1658	2.79 [1.29]	-0.37*** (0.11)	0.02 (0.09)	0.11 (0.15)

Additional results from ???. All variables measuring values are in USD. Results are qualitatively unchanged when using logs. Indices are constructed as the sum of normalized components, and are then normalized to have mean 0 and SD 1 for control respondents in the baseline. In all the regressions, we control for respondent age, gender, education, whether the meter is paid directly by the user, number of meter users, whether the respondent is part of the household or business sample, whether the location includes both a household and a business, and district fixed effects. The control mean is the mean for control sites in the baseline period. Standard errors are clustered at the site level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$